PRODUCT USER MANUAL

For Sea Level Altimeter products

SEALEVEL_GLO_PHY_L4_NRT_ 008_046
SEALEVEL_EUR_PHY_L4_NRT_ 008_060
SEALEVEL_GLO_PHY_L4_MY_008_047
SEALEVEL_EUR_PHY_L4_MY_008_068
Issue: 1.0

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Approval Date : November 2025

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/05/28		Change Format file: subdivision of unique PUM for L3/L4 products in 3 documents. This new doc concern L4 products New NRT demo dataset : NRT L4 with SWOT-KaRIn New MY demo dataset : Reprocessed L4 with enhanced MIOST technique	K Grassi	M-I Pujol

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

The Sea Level TAC (Thematic Assembly Centre) is one of the eight TAC of the Copernicus Marine Service. The aim of this document is to describe the products delivered by the Sea Level TAC.
The data produced in the frame of this TAC are generated by the processing system including data from all altimeter Copernicus missions (Sentinel-6A, Sentinel-3A/B) and other collaborative or opportunity missions (e.g. :Jason-3, Saral[-DP]/AltiKa, Cryosat-2, OSTM/Jason-2, Jason-1, Topex/Poseidon, Envisat, GFO, ERS-1/2, Haiyang-2A/B).
The SL-TAC main products lines consist in Level-3 (L3; along-track) and Level-4 (L4; gridded) altimeter products. Both are produced in Near Real Time (NRT) or Delayed Time/ Multi-Year (DT/MY). Through the SL-TAC, the Copernicus Marine Service also disseminates L3 products/datasets produced under the Eumetsat responsibility. They mainly concern the Copernicus altimeter mission (Sentinel-6). Additionally, auxiliary/statics product can be delivered for additional information on the L3/L4 products quality. We describe here after the different product L4 lines.
The data provided to users have a global coverage and regional products are also computed over European Seas, including part of the North-East Atlantic, Baltic, Mediterranean and Black Seas.

a. Near Real time products

The user can check the version of the product used with the global attributes given in netCDF files:
product_version : gives the information of the main Copernicus Marine release for which the last change has been implemented (e.g. "vDec2021")
software_version : gives the precise software version used for the production, and information of the reference altimeter standards and processing used (e.g. : "19.0.0_DUACS_DT2021_baseline")
The system's noticeable events, outage or changes, as well as details of the impact on product quality events can be found in the associated QUID document (see section V)

b. Multi-Year products

The user can check the version of the product used with the global attributes given in netCDF files:
product_version : gives the information of the main release for which the last change has been implemented (e.g. "vNov2024")
software_version : gives the precise software version used for the production, and information of the reference altimeter standards and processing used (e.g. : "8.0_DUACS_DT2024_baseline")
The lists the system's noticeable events, outage or changes, as well as details of the impact on product quality events can be found in the associated QUID document (see section "V.4: V.3 REP/DT sub-system version changes").

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SEALEVEL_GLO_PHY_L4_NRT _008_046 SEALEVEL_EUR_PHY_L4_NRT _008_060 SEALEVEL_GLO_PHY_L4_MY_008_047 SEALEVEL_EUR_PHY_L4_MY_008_068
Geographical coverage	GLO : Global Ocean EUR : European Area (30.0625°W-42.0625°E ; 19.9375°N-66.0625°N)
Variables	The different variable included in this product are listed in section 1.2.2
Product Type	NRT : Near Real Time MY : Multi Year
Available time series	NRT : usually the last 2 years are available online for historic product/dataset lines. This period can be reduced at the start of production of a new product/dataset line MY : 1993/01/01 ; -5 to 9M before present.
Temporal resolution	Daily
Update Frequency	Daily
Target delivery time	NRT: daily MY : ~4 times a year
Delivery mechanism	Copernicus Marine Information System
Horizontal resolution	GLO : 0.125°x0.125° EUR : 0. 0625° x 0.0625°
Number of vertical levels	1 (sea surface)
Format	Netcdf CF1.6

b) Production System Description

The SEALEVEL altimeter products are processed with the CNES/CLS DUACS (Data Unification and Altimeter Combination System) system (https://duacs.cls.fr/)
The DUACS processing steps are described in the QUID document.

The altimeter measurements used as input of the DUACS system consist in a majority of Level 2 products from different missions, that are available under three forms, with different delay of availability:

- Fast delivery or Near Real Time products (OGDR or NRT). These products do not always benefit from precise orbit determination, nor from some external model-based corrections (Dynamic Atmospheric Correction (DAC), Global Ionospheric Maps (GIM)).
- The Intermediate or Slow Time Critical products (IGDR or STC) that are the latest high-quality altimeter data produced in near-real-time
- Delayed Time or Non-Time Critical product (GDR or NTC).

Details of the different L2 altimeter products sources and delay of availability are given in the QUID document.

c) Dataset details

c. L4 NRT datasets

SEALEVEL_GLO_PHY_L4_NRT_008_046	**SEALEVEL_EUR_PHY_L4_NRT_008_060**
Dataset: cmems_obs-sl_glo_phy-ssh_nrt_allsat-l4-duacs-0.125deg _P1D cmems_obs-sl_glo_phy-ssh_nrt_demo-allsat-swos-l4-duacs-0.125deg_P1D-i*	**Dataset:** cmems_obs-sl_eur_phy-ssh_nrt_allsat-l4-duacs-0.0625deg _P1D cmems_obs-sl_eur_phy-ssh_nrt_demo-allsat-swos-l4-duacs-0.0625deg_P1D-i *
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name	
sla [m] Sea level anomaly sea_surface_height_above_sea_level	
err_sla [m] Formal mapping error sea_surface_height_above_sea_level standard_error	
ugosa [m/s] Geostrophic velocity anomalies: zonal component surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid	
err_ugosa [m/s] Formal mapping error on zonal geostrophic velocity anomalies surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid standard_error	
vgosa [m/s] Geostrophic velocity anomalies: meridian component surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid	
err_vgosa [m/s] Formal mapping error on meridional geostrophic velocity anomalies surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid standard_error	

adt [m] Absolute dynamic topography sea_surface_height_above_geoid
ugos [m/s] Absolute geostrophic velocity: zonal component surface_geostrophic_eastward_sea_water_velocity
vgos [m/s] Absolute geostrophic velocity: meridian component surface_geostrophic_northward_sea_water_velocity
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name
time [days since 1950-01-01 00:00:00] Time of measurement time
longitude [degrees_east] Longitude of measurement longitude
latitude [degrees_north] Latitude of measurement latitude
lon_bnds [degrees_east] - -
Lat_bnds[degrees_north] - -
Other useful variables name in the NetCDF file and Unit: Long_name & Standard_name
flag_ice [0,1] Ice Flag for a 15% criterion of ice concentration status_flag

**merging the nadir measurements with the SWOT-KaRIn. Due to the risk inherent to the "non operational status" of the required L3 KaRIn product, it's a demonstrative dataset*

Table 1 : List of the datasets and variable names and unit for the SEALEVEL_GLO_PHY_L4_NRT_008_046 and SEALEVEL_EUR_PHY_L4_NRT_008_060 products

d) L4 MY datasets

SEALEVEL_GLO_PHY_L4_MY_008_047	SEALEVEL_EUR_PHY_L4_MY_008_068
Dataset: cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.125deg_P1D cmems_obs-sl_glo_phy-ssh_my_allsat-demo-l4-duacs-0.125deg_P1D-i*	**Dataset:** cmems_obs-sl_eur_phy-ssh_my_allsat-l4-duacs-0.0625deg_P1D cmems_obs-sl_eur_phy-ssh_my_allsat-demo-l4-duacs-0.0625deg_P1D-i*
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name	
sla [m] Sea level anomaly sea_surface_height_above_sea_level	
err_sla [m] Formal mapping error	

sea_surface_height_above_sea_level standard_error	
ugosa [m/s] Geostrophic velocity anomalies: zonal component surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid	
err_ugosa [m/s] Formal mapping error on zonal geostrophic velocity anomalies surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid standard_error	
vgosa [m/s] Geostrophic velocity anomalies: meridian component surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid	
err_vgosa [m/s] Formal mapping error on meridional geostrophic velocity anomalies surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid standard_error	
adt [m] Absolute dynamic topography sea_surface_height_above_geoid	
ugos [m/s] Absolute geostrophic velocity: zonal component surface_geostrophic_eastward_sea_water_velocity	
vgos [m/s] Absolute geostrophic velocity: meridian component surface_geostrophic_northward_sea_water_velocity	
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name	
time [days since 1950-01-01 00:00:00] Time of measurement time	
longitude [degrees_east] Longitude of measurement longitude	
latitude [degrees_north] Latitude of measurement latitude	
lon_bnds [degrees_east] - -	
Lat_bnds[degrees_north] - -	
Other useful variables name in the NetCDF file and Unit: Long_name & Standard_name	
flag_ice [0,1] Ice Flag for a 15% criterion of ice concentration status_flag	
tpa_correction [m] TOPEX-A instrumental drift correction derived from altimetry and tide gauges global comparisons (WCRP Sea Level Budget Group, 2018) sea_surface_height_above_sea_level	
Dataset: cmems_obs-sl_glo_phy-	Dataset: cmems_obs-sl_eur_phy-

ssh_my_allsat-l4-duacs-0.25deg125deg_P1M-m	ssh_my_allsat-l4-duacs-0.0625deg_P1M-m
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name	
sla [m] Sea level anomaly sea_surface_height_above_sea_level	
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name	
time [days since 1950-01-01 00:00:00] Time of measurement time	
longitude [degrees_east] Longitude of measurement longitude	
latitude [degrees_north] Latitude of measurement latitude	
lon_bnds [degrees_east] - -	
Lat_bnds[degrees_north] - -	

**30y reprocessing with *MIOST version. It's a demontrative production that will not be extended in time*

Table 2 : List of the datasets and variable names and unit for the SEALEVEL_GLO_PHY_L4_MY_008_047 and SEALEVEL_EUR_PHY_L4_MY_008_068 products

e) Additional information on some parameters

Sla [m] **sla_filtered** **sla_unfiltered**	Sea Level Anomaly (see Annex) Unfiltered : raw measurement including noises Filtered: low-pass filtered SLA for noise reduction
err_sla [m] **err_ugosa |m/s]** **err_vgosa [m/s]**	Formal mapping error (see QUID document) associated to sla, ugosa and vgosa variables
ugosa & vgosa [m/s] **ugos & vgos [m/s]**	Anomalies of the geostrophic current and absolute geostrophic current components (see QUID document) in the zonal and meridian directions
Adt [m]	Absolute Dynamic Topography (see Annex)

f) Processing information

The NRT L4 products are disseminated once a day, nominally before 4 am UTC.
The MY products are extended in time nearly 3 or 4 times per years, depending on the upstream data availability.

g) Catalogue updating

d. Update Time for Near real time production: Level-4 gridded products

The exploitation of near-real-time and fast-delivery data allows the DUACS system to produce multi-mission maps with 0-day, 3-day and 6-day delay, using non-centered processing time-window (in NRT case, "future" data are not available; the computation time window takes into account only the 6 weeks before the date).

Those products are delivered every day.

Three merged maps are produced daily, each with a different delay and quality:

- A 6-day delay, which represents a final NRT map production,
- A 3-day delay, which represents an intermediate map production,
- and a 0-day delay, which represents a preliminary map production, based on IGDR+OGDR production.

Then, thesel ntermediate and preliminary maps are replaced when a better quality data is available:

- At d0+6, the final NRT map replaces the preliminary map which was produced at d0.

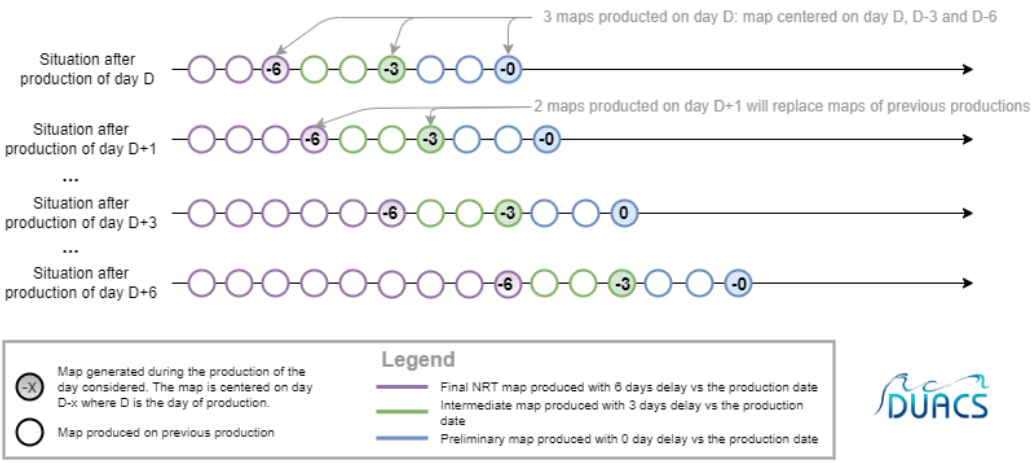

Figure 2: Three merged maps are produced daily: final map (d-6), intermediate map (d-3) and preliminary map (d0)

e. Update Time for Multi-Year production

The products/datasets are extended in time nearly four times a year in the nominal case, so that these datasets cover from 1993 up to five to nine months before present. These datasets benefit from the highest quality altimeter measurements and geophysical corrections, and are produced with a unique system (no changes in time in the set up, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.

A full reprocessing of the Multi-Year series is usually delivery each 3 to 4 years

h) Time averaging

L4 monthly mean dataset is obtained by simple temporal average of the L4 daily products available over the month considered.

i) Product use and impact

The L4 DUACS products can be used for different applications. We give here some examples:

- Monitoring of the mesoscale activity: different algorithms of eddy detection and tracking can be applied using sea level given in the L4 DUACS products (e.g. META products)
- Scientific studies regarding physical and biogeochemical domains (see for instance DUACS studies and applications)
- Applications that can be defined using sea level and/or surface currents provided in the products (e.g. boat route optimization; frontal zone detection and monitoring; etc.). Note however that the DUACS L4 products provide only the geostrophic component of the surface current. Other models and multi-observations products available on the Copernicus Marine Service catalog provide the total surface currents including ageostrophic components.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

The scheme is: **datasetname_nnnnnn.nc**

where :

- **datasetname** is a character string within one of the datasets listed in section I.2.2 and I.2.3 :
- **nnnnnn**: 6digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **nc**: standard NetCDF filename extension.

Example:

cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.125deg_P1D_1644250104522.nc

Nomenclature of original files: Level-4 gridded products

daily datasets

The nomenclature used is:

<delay>_<zone>_allsat_phy_l4_<datemap>_<dateprod>.<format>

where the fields in "<>" are described below:

delay	nrt dt	near-real time products delayed time products
zone	global europe	global geographic coverage product Europe products
date	YYYYMMDD	date of the map

dateprod	YYYYMMDD	production date of the map
format	.nc	NetCdf CF1.6

Monthly mean datasets

The nomenclature used is:

<delay>_<zone>_allsat_msla_h_y<YYYY>_m<MM>.<format>

where the fields in "<>" are described below:

delay	dt	delayed time products
zone	global	global geographic coverage product
date	YYYY MM	year of the map month of the map
format	.nc	NetCdf CF1.6

4.1) FILE SIZE

The following table give an estimation of the typical size of one the daily file for L3 and L4 products, when downloaded through Marine Data Store.

Area	Product Level	NetCDF file size
Global	L4	30 MB
Europe	L4	5MB

4.2) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**.

5) REFERENCES

Quality Information Document CMEMS-SL-QUID-008-032-068.pdf

6) ANNEX

An example output NetCDF file header for the cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.25deg_P1D dataset is given bellow.

```
netcdf dt_global_allsat_phy_l4_20200101_20210726 {
dimensions:
	time = 1 ;
	latitude = 720 ;
	longitude = 1440 ;
	nv = 2 ;
variables:
	int crs ;
		crs:comment = "This is a container variable that describes the grid_mapping used by the data in this file. This variable does not contain any data; only information about the geographic coordinate system." ;
		crs:grid_mapping_name = "latitude_longitude" ;
		crs:inverse_flattening = 298.257 ;
		crs:semi_major_axis = 6378136.3 ;
	float time(time) ;
		time:axis = "T" ;
		time:calendar = "gregorian" ;
		time:long_name = "Time" ;
		time:standard_name = "time" ;
		time:units = "days since 1950-01-01 00:00:00" ;
	float latitude(latitude) ;
		latitude:axis = "Y" ;
		latitude:bounds = "lat_bnds" ;
		latitude:long_name = "Latitude" ;
		latitude:standard_name = "latitude" ;
		latitude:units = "degrees_north" ;
		latitude:valid_max = 89.875 ;
		latitude:valid_min = -89.875 ;
	float lat_bnds(latitude, nv) ;
		lat_bnds:comment = "latitude values at the north and south bounds of each pixel." ;
		lat_bnds:units = "degrees_north" ;
	float longitude(longitude) ;
		longitude:axis = "X" ;
		longitude:bounds = "lon_bnds" ;
		longitude:long_name = "Longitude" ;
		longitude:standard_name = "longitude" ;
		longitude:units = "degrees_east" ;
		longitude:valid_max = 179.875 ;
		longitude:valid_min = -179.875 ;
	float lon_bnds(longitude, nv) ;
		lon_bnds:comment = "longitude values at the west and east bounds of each pixel." ;
		lon_bnds:units = "degrees_east" ;
	int nv(nv) ;
		nv:comment = "Vertex" ;
		nv:long_name = "Number of cell vertices" ;
		nv:units = "1" ;
	int sla(time, latitude, longitude) ;
		sla:_FillValue = -2147483647 ;
		sla:ancillary_variables = "err_sla" ;
		sla:comment = "The sea level anomaly is the sea surface height above mean sea surface; it is referenced to the [1993, 2012] period; see the product user manual for details" ;
		sla:coordinates = "longitude latitude" ;
		sla:grid_mapping = "crs" ;
		sla:long_name = "Sea level anomaly" ;
		sla:scale_factor = 0.0001 ;
		sla:standard_name = "sea_surface_height_above_sea_level" ;
		sla:units = "m" ;
	int err_sla(time, latitude, longitude) ;
		err_sla:_FillValue = -2147483647 ;
		err_sla:comment = "The formal mapping error represents a purely theoretical mapping error. It mainly traduces errors induced by the constellation sampling capability and consistency with the spatial/temporal scales considered, as described in Le Traon et al (1998) or Ducet et al (2000)" ;
		err_sla:coordinates = "longitude latitude" ;
		err_sla:grid_mapping = "crs" ;
		err_sla:long_name = "Formal mapping error" ;
		err_sla:scale_factor = 0.0001 ;
		err_sla:standard_name = "sea_surface_height_above_sea_level standard_error" ;
```

```
            err_sla:units = "m" ;
      int ugosa(time, latitude, longitude) ;
            ugosa:_FillValue = -2147483647 ;
            ugosa:ancillary_variables = "err_ugosa" ;
            ugosa:comment = "The geostrophic velocity anomalies are referenced to the [1993, 2012] period" ;
            ugosa:coordinates = "longitude latitude" ;
            ugosa:grid_mapping = "crs" ;
            ugosa:long_name = "Geostrophic velocity anomalies: zonal component" ;
            ugosa:scale_factor = 0.0001 ;
            ugosa:standard_name = "surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid" ;
            ugosa:units = "m/s" ;
      int err_ugosa(time, latitude, longitude) ;
            err_ugosa:_FillValue = -2147483647 ;
            err_ugosa:comment = "The formal mapping error represents a purely theoretical mapping error. It mainly traduces
errors induced by the constellation sampling capability and consistency with the spatial/temporal scales considered, as described
in Le Traon et al (1998) or Ducet et al (2000)" ;
            err_ugosa:coordinates = "longitude latitude" ;
            err_ugosa:grid_mapping = "crs" ;
            err_ugosa:long_name = "Formal mapping error on zonal geostrophic velocity anomalies" ;
            err_ugosa:scale_factor = 0.0001 ;
            err_ugosa:standard_name = "surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid
standard_error" ;
            err_ugosa:units = "m/s" ;
      int vgosa(time, latitude, longitude) ;
            vgosa:_FillValue = -2147483647 ;
            vgosa:ancillary_variables = "err_vgosa" ;
            vgosa:comment = "The geostrophic velocity anomalies are referenced to the [1993, 2012] period" ;
            vgosa:coordinates = "longitude latitude" ;
            vgosa:grid_mapping = "crs" ;
            vgosa:long_name = "Geostrophic velocity anomalies: meridian component" ;
            vgosa:scale_factor = 0.0001 ;
            vgosa:standard_name = "surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid" ;
            vgosa:units = "m/s" ;
      int err_vgosa(time, latitude, longitude) ;
            err_vgosa:_FillValue = -2147483647 ;
            err_vgosa:comment = "The formal mapping error represents a purely theoretical mapping error. It mainly traduces
errors induced by the constellation sampling capability and consistency with the spatial/temporal scales considered, as described
in Le Traon et al (1998) or Ducet et al (2000)" ;
            err_vgosa:coordinates = "longitude latitude" ;
            err_vgosa:grid_mapping = "crs" ;
            err_vgosa:long_name = "Formal mapping error on meridional geostrophic velocity anomalies" ;
            err_vgosa:scale_factor = 0.0001 ;
            err_vgosa:standard_name = "surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid
standard_error" ;
            err_vgosa:units = "m/s" ;
      int adt(time, latitude, longitude) ;
            adt:_FillValue = -2147483647 ;
            adt:comment = "The absolute dynamic topography is the sea surface height above geoid; the adt is obtained as follows:
adt=sla+mdt where mdt is the mean dynamic topography; see the product user manual for details" ;
            adt:coordinates = "longitude latitude" ;
            adt:grid_mapping = "crs" ;
            adt:long_name = "Absolute dynamic topography" ;
            adt:scale_factor = 0.0001 ;
            adt:standard_name = "sea_surface_height_above_geoid" ;
            adt:units = "m" ;
      int ugos(time, latitude, longitude) ;
            ugos:_FillValue = -2147483647 ;
            ugos:coordinates = "longitude latitude" ;
            ugos:grid_mapping = "crs" ;
            ugos:long_name = "Absolute geostrophic velocity: zonal component" ;
            ugos:scale_factor = 0.0001 ;
            ugos:standard_name = "surface_geostrophic_eastward_sea_water_velocity" ;
            ugos:units = "m/s" ;
      int vgos(time, latitude, longitude) ;
            vgos:_FillValue = -2147483647 ;
            vgos:coordinates = "longitude latitude" ;
            vgos:grid_mapping = "crs" ;
            vgos:long_name = "Absolute geostrophic velocity: meridian component" ;
            vgos:scale_factor = 0.0001 ;
            vgos:standard_name = "surface_geostrophic_northward_sea_water_velocity" ;
            vgos:units = "m/s" ;
      int flag_ice(time, latitude, longitude) ;
            flag_ice:_FillValue = -2147483647 ;
```

```
        flag_ice:comment = "Ice Flag based on CDR OSI SAF products until 2016 (OSI-450), Interim products from 2016 (OSI-430-b) (Lavergne et al., 2019). The flag corresponds to the 15% sea ice concentration." ;
        flag_ice:long_name = "Ice Flag for a 15% criterion of ice concentration" ;
        flag_ice:standard_name = "status_flag" ;
        flag_ice:flag_values = 0, 1 ;
        flag_ice:flag_meanings = "data_on_sea data_on_ice" ;
        flag_ice:grid_mapping = "crs" ;
        flag_ice:coordinates = "longitude latitude" ;
    int tpa_correction(time) ;
        tpa_correction:_FillValue = -2147483647 ;
        tpa_correction:comment = "This variable can be added to the gridded SLA to correct for the observed instrumental drift during the lifetime of the TOPEX-A mission (the correction is null after this period). This is a global correction to be added a posteriori (and not before) on the global mean sea level estimate derived from the gridded sea level map. It can be applied at regional or local scale as a best estimate (better than no correction, since the regional variation of the instrumental drift is unknown). See product manual for more details." ;
        tpa_correction:long_name = "TOPEX-A instrumental drift correction derived from altimetry and tide gauges global comparisons (WCRP Sea Level Budget Group, 2018)" ;
        tpa_correction:scale_factor = 0.0001 ;
        tpa_correction:standard_name = "sea_surface_height_above_sea_level" ;
        tpa_correction:units = "m" ;

// global attributes:
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :cdm_data_type = "Grid" ;
        :comment = "Sea Surface Height measured by Altimetry and derived variables" ;
        :contact = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_name = "CMEMS - Sea Level Thematic Assembly Center" ;
        :creator_url = "http://marine.copernicus.eu" ;
        :date_created = "2021-07-27T15:26:37Z" ;
        :date_issued = "2021-07-27T15:26:37Z" ;
        :date_modified = "2021-07-27T15:26:37Z" ;
        :geospatial_lat_max = 89.875 ;
        :geospatial_lat_min = -89.875 ;
        :geospatial_lat_resolution = 0.25 ;
        :geospatial_lat_units = "degrees_north" ;
        :geospatial_lon_max = 179.875 ;
        :geospatial_lon_min = -179.875 ;
        :geospatial_lon_resolution = 0.25 ;
        :geospatial_lon_units = "degrees_east" ;
        :geospatial_vertical_max = 0. ;
        :geospatial_vertical_min = 0. ;
        :geospatial_vertical_positive = "down" ;
        :geospatial_vertical_resolution = "point" ;
        :geospatial_vertical_units = "m" ;
        :history = "2021-07-27 15:26:39Z: Creation" ;
        :institution = "CLS, CNES" ;
        :keywords = "Oceans > Ocean Topography > Sea Surface Height" ;
        :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
        :platform = Altika Drifting Phase, Cryosat-2, Haiyang-2A Geodetic Phase, Haiyang-2B, Jason-3, Sentinel-3A, Sentinel-3B";
        :processing_level = "L4" ;
        :product_version = "vDec2021" ;
        :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
        :references = "http://marine.copernicus.eu" ;
        :software_version = "7.0_DUACS_DT2021_baseline" ;
        :source = "Altimetry measurements" ;
        :ssalto_duacs_comment = "The reference mission used for the altimeter inter-calibration processing is Topex/Poseidon between 1993-01-01 and 2002-04-23, Jason-1 between 2002-04-24 and 2008-10-18, OSTM/Jason-2 between 2008-10-19 and 2016-06-25, Jason-3 since 2016-06-25." ;
        :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
        :summary = "SSALTO/DUACS Delayed-Time Level-4 sea surface height and derived variables measured by multi-satellite altimetry observations over Global Ocean." ;
        :time_coverage_duration = "P1D" ;
        :time_coverage_end = "2020-01-01T12:00:00Z" ;
        :time_coverage_resolution = "P1D" ;
        :time_coverage_start = "2019-12-31T12:00:00Z" ;
        :title = "DT merged all satellites Global Ocean Gridded SSALTO/DUACS Sea Surface Height L4 product and derived variables" ;
}
```